

Dear Editors:

We would like to submit the enclosed manuscript entitled “The Vanishing First Rung: Generative Artificial Intelligence, Entry-Task Exposure and Youth Employment in Chinese Listed Firms” for consideration in *Systems*, for the Special Issue “Systems Approaches to Generative AI: Workforce Development, Organisational Learning, and Economic Transformation”. The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by all authors.

Generative artificial intelligence is unusual among labour-saving technologies in that the work it performs best is the work through which people become qualified for the work it cannot perform. Drafting, reconciliation, summarising and first-pass analysis are simultaneously a firm’s production input and its training input. A firm that automates them captures the saving immediately and pays for the lost learning years later, in a currency — experienced staff — that cannot be bought back quickly. This manuscript asks whether that coincidence has already begun to close the entry rung of the firm-internal job ladder, and answers it with a design built around the late-2022 arrival of general-purpose language-model assistants: a shock that reached every firm at the same moment but bit in proportion to how much of its entry-level work lay inside the newly automatable frontier.

The highlights of the paper are as follows.

(1) A displacement that lands on one age margin and not the others. Using 31,015 firm-year observations for 3,240 Chinese A-share listed firms over 2014–2024, a one-standard-deviation increase in pre-shock entry-task exposure lowers the youth employment share by 1.50 percentage points, 5.42 per cent of its mean. The five pre-shock leads are individually and jointly insignificant ($\chi^2 = 4.73$, $p = 0.450$), the effect appears in the shock year and persists, and it survives instrumental variables, propensity-score matching, entropy balancing and eight further specifications. The placebo is decisive: run on employees aged 41 to 50, whose tasks lie outside the exposed frontier, the coefficient is -0.0011 and insignificant, two orders of magnitude below the baseline.

(2) The channel that works, and the channel that is assumed to work and does not. About 31 per cent of the effect travels through the observable contraction of entry-level routine cognitive positions (indirect effect -0.0368 , bootstrap confidence interval $[-0.0441, -0.0301]$). Exposed firms do raise training expenditure, significantly so, and the offsetting indirect effect is nonetheless 0.0026 with a confidence interval containing zero. The compensating mechanism that most policy discussion takes for granted is present in intention and absent in effect.

(3) What protects the rung is capability, not incentive. Organisational learning capacity significantly attenuates the loss; internal labour-market depth, which gives a firm every reason to want its entry rung protected, does not. Firms that already know how to build competence deliberately can afford to lose the rung that used to build it for them; firms that merely need the competence cannot. The reduction in youth employment is followed by a fall in internal advancement, about half of which runs through the contracted entry rung itself.

We believe the manuscript is well suited to this Special Issue and to *Systems*. Its contribution is not the decoration of an economic result with systems vocabulary but the identification of a specific, estimable instance of the classical socio-technical proposition that partial optimisation degrades the whole: generative artificial intelligence raises the productivity of the technical subsystem while quietly degrading the reproduction function of the social subsystem, because the tasks it absorbs belong to both. The practical payoff is that the diagnostic quantity — the share of entry-grade positions that still contain graded, supervised, consequential work — is observable now, from disclosures firms already file, whereas the capability consequence it foreshadows becomes observable only once it is expensive to reverse. The urgency is not hypothetical: youth unemployment remains far above the adult rate worldwide, and the earnings penalty for a poor labour-market entry is measured in decades. We deeply appreciate your consideration of our manuscript and look forward to receiving comments from the reviewers. Correspondence should be directed to the address below.

Thanks very much for your attention to our paper.

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Very sincerely yours,

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